

Backend Testing Guide – AI Complaint Management System

Comprehensive checklist for verifying APIs, database operations, authentication, AI services, security, notifications, and performance.

1. Authentication Module

- Employee Login: Verify valid credentials return JWT token and role.
- Invalid credentials return 401 Unauthorized.
- Admin and HR logins return correct roles and permissions.

2. Complaint Creation

- Verify complaint is saved in database.
- Status defaults to Open.
- Complaint ID and timestamps are generated.

3. AI Categorization

- Verify complaint text is categorized correctly.
- Test Payroll, HR, Leave Management, and IT Support categories.

4. AI Priority Assignment

- Low, Medium, and High priorities assigned correctly.

5. Duplicate Detection

- Verify similar complaints are identified and linked.

6. Complaint Listing API

- Pagination, filtering, and sorting work correctly.

7. Complaint Details API

- Returns complete complaint information.

8. Complaint Assignment

- Admin can assign complaints to HR.
- Assignment history is stored.

9. Status Updates

- HR can change status: Open, In Progress, Resolved, Closed.

10. Complaint Timeline

- Audit trail records all major events.

11. Notifications

- Notifications generated for creation, assignment, and resolution.

12. Role-Based Access Control

- Employee, HR, and Admin permissions verified.

13. Dashboard APIs

- Dashboard counts match database values.

14. Database Validation

- Verify complaints and users data integrity.

15. API Failure Testing

- 401 Unauthorized, 404 Not Found, and 500 Server Error scenarios.

16. Performance Testing

- Complaint creation <500ms.
- Complaint fetch <300ms.
- AI classification <2s.

Postman Testing Checklist

- Authentication APIs
- Complaint APIs
- AI Services
- Management APIs
- Security Tests
- Database Verification